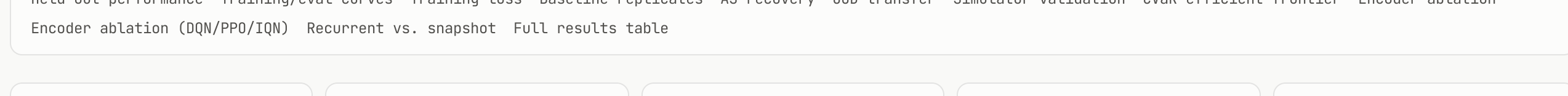


DistRL Market Maker – Training & Evaluation Results

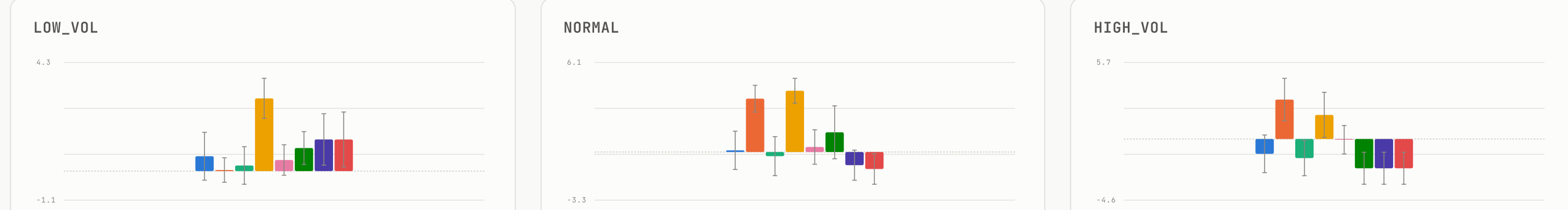
5 RL agents vs. 3 analytical baselines across 3 volatility regimes in an ABIDES-Gym limit-order-book simulator, plus out-of-distribution transfer, CVar risk-sensitivity, and state-representation ablations.



RL RUNS	EARLY STOPPED	BEST HELD-OUT SHARPE (RL)	BEST HELD-OUT SHARPE (BASELINE)	REGIMES WHERE AN RL AGENT BEATS GLTF
15 5 agents × 3 regimes	9 / 15 1000 episodes saved (13%)	+4.194 QR-DQN - normal - 15 unseen episodes	+1.348 Fixed-Spread - normal	3 / 3 on held-out data, GLTF is the strictest reference

Held-out test performance – the real comparison

Each model's best checkpoint (or fixed policy) evaluated on 15 held-out episodes never touched by training or selection, with every model seeing the identical market conditions per regime. This is the headline result — everything below is a training-time diagnostic.

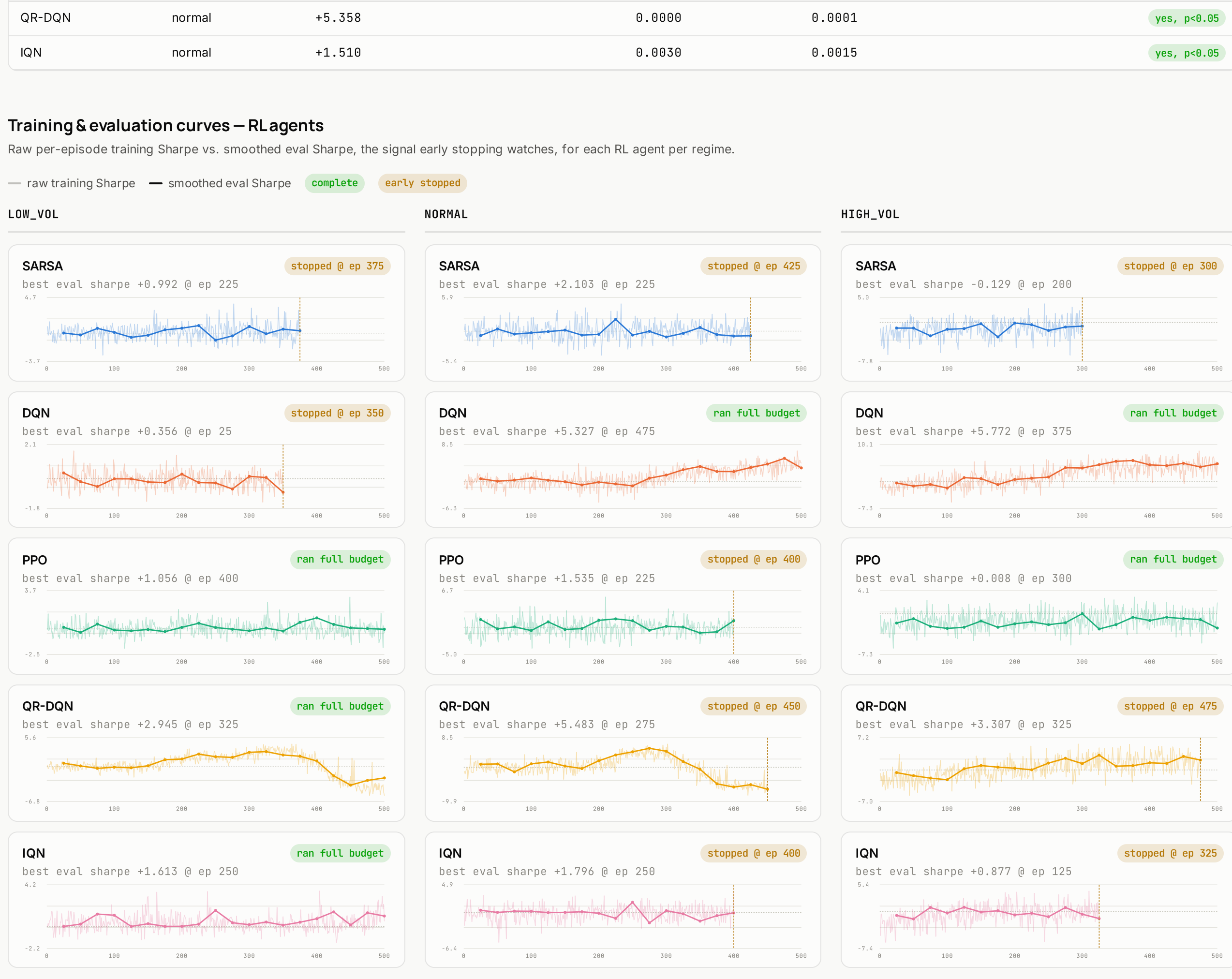


Paired significance test (t-test, Wilcoxon) of each RL agent vs. GLTF, the strictest classical baseline, on the same 15 shared held-out episodes.

AGENT	REGIME	MEAN Δ VS. GLTF	T-TEST P	WILCOXON P	SIGNIFICANTLY BEATS GLTF?
SARSA	high_vol	+1.075	0.8153	0.0256	yes, p<0.05
DQN	high_vol	+5.112	0.0000	0.0001	yes, p<0.05
PPO	high_vol	+0.759	0.1308	0.2293	no
QR-DQN	high_vol	+3.971	0.0000	0.0001	yes, p<0.05
IQN	high_vol	+2.125	0.0001	0.0002	yes, p<0.05
SARSA	low_vol	-0.657	0.0799	0.0946	no
DQN	low_vol	-1.199	0.0029	0.0034	no
PPO	low_vol	-1.021	0.0229	0.0215	no
QR-DQN	low_vol	+1.613	0.0000	0.0001	yes, p<0.05
IQN	low_vol	-0.806	0.0036	0.0043	no
SARSA	normal	+1.278	0.0003	0.0001	yes, p<0.05
DQN	normal	+4.818	0.0000	0.0001	yes, p<0.05
PPO	normal	+0.880	0.0614	0.0151	no
QR-DQN	normal	+5.358	0.0000	0.0001	yes, p<0.05
IQN	normal	+1.510	0.0030	0.0015	yes, p<0.05

Training & evaluation curves – RL agents

Raw per-episode training Sharpe vs. smoothed eval Sharpe, the signal early stopping watches, for each RL agent per regime.



Training loss / objective – RL agents

Each agent's own training-objective loss on a log scale — not comparable across algorithms, since each optimizes a different loss function.

Legend: raw per-episode loss (blue), 15-episode rolling mean (orange).



Baseline replicate performance

Performance variance across 50 independent replicates of each fixed (non-learning) baseline policy, per regime.



AS-recovery: does the agent rediscover analytical inventory-skew quoting?

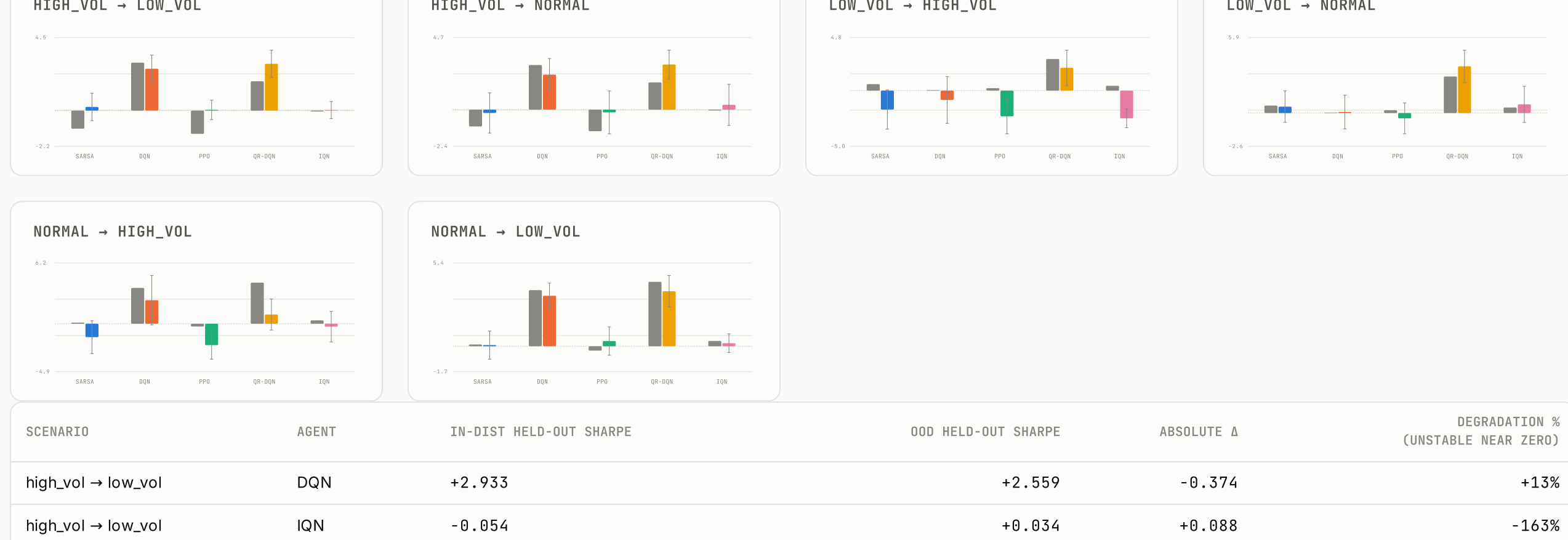
How closely each agent's inventory-skew quoting matches the closed-form AS/GLTF curves (R²) — a sanity check on learned behavior, not a target to beat.



OOD transfer: agents evaluated outside their training regime

Each agent's performance from one regime dropped into a different regime with no fine-tuning, across all 6 directed pairs among low_vol/normal/high_vol — tests generalization to market conditions never seen in training. QR-DQN stays profitable in all 6 of 8 transfers and actually improves on its in-distribution Sharpe in 3 of them; transferring into high_vol is the hardest direction for most agents.

Legend: in-distribution (blue), out-of-distribution (orange).



SCENARIO	AGENT	IN-DIST. HELD-OUT SHARPE	OOD HELD-OUT SHARPE	ABSOLUTE Δ	DEGRADATION % (UNSTABLE NEAR ZERO)
high_vol → low_vol	DQN	+2.933	+2.559	-0.374	+13%
high_vol → low_vol	DQN	-0.054	-0.054	0.000	-163%
high_vol → low_vol	QR-DQN	+1.793	+2.862	+1.069	-68%
high_vol → low_vol	SARSA	-1.104	+0.224	+1.328	-128%
high_vol → low_vol	PPO	-1.419	+0.046	+1.465	-103%
high_vol → normal	DQN	+2.933	+2.300	-0.633	+22%
high_vol → normal	IQN	-0.054	+0.313	+0.368	-678%
high_vol → normal	SARSA	-1.104	-0.220	+0.884	-80%
high_vol → normal	QR-DQN	+1.793	+2.970	+1.178	-66%
high_vol → normal	PPO	-1.419	-0.181	+1.238	-87%
low_vol → high_vol	IQN	+0.434	-2.505	-2.939	+677%
low_vol → high_vol	PPO	+0.219	-2.320	-2.539	+1159%
low_vol → high_vol	SARSA	+0.583	-1.710	-2.293	+393%
low_vol → high_vol	DQN	+0.041	-0.845	-0.886	n/a (in-dist = 0)
low_vol → high_vol	QR-DQN	+2.053	+2.059	-0.794	+28%
low_vol → normal	PPO	+0.219	-0.411	-0.630	+288%
low_vol → normal	SARSA	+0.583	+0.502	-0.081	+14%
low_vol → normal	DQN	+0.041	+0.082	+0.041	n/a (in-dist = 0)
low_vol → normal	IQN	+0.434	+0.681	+0.247	+5%
low_vol → normal	QR-DQN	+2.053	+3.651	+0.798	-28%
normal → high_vol	PPO	+4.194	+0.939	-3.255	+78%
normal → high_vol	DQN	-0.284	-2.184	-1.900	+668%
normal → high_vol	SARSA	+0.114	-1.370	-1.484	+1307%
normal → high_vol	IQN	+3.653	+2.400	-1.253	+34%
normal → high_vol	QR-DQN	+0.345	-0.298	-0.643	+186%
normal → low_vol	QR-DQN	+4.194	+3.581	-0.613	+15%
normal → low_vol	DQN	+3.653	+3.286	-0.367	+10%
normal → low_vol	IQN	+0.345	+0.197	-0.148	+43%
normal → low_vol	SARSA	+0.114	+0.075	-0.039	+34%
normal → low_vol	PPO	-0.284	-0.338	-0.622	-21%

Simulator validation – stylized facts

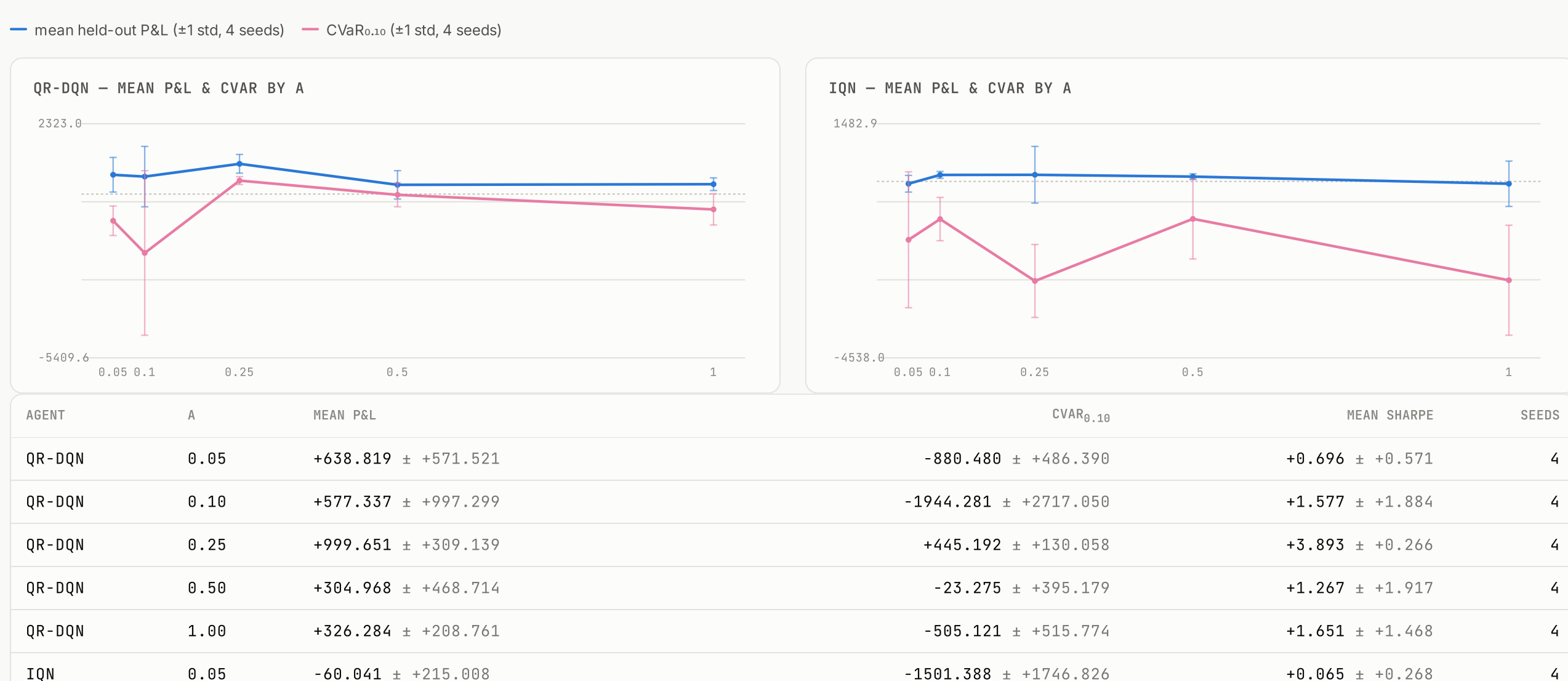
A random-policy audit of the simulator itself against 5 standard empirical market-microstructure regularities, independent of any trained agent — validates that the environment produces realistic dynamics to learn from.



CVar efficient frontier – QR-DQN & IQN, normal regime

Mean P&L vs. tail risk (CVar_{0.01}) across the CVar-α sweep, averaged over 4 training seeds per setting. Most α values are statistically indistinguishable once seed variance is accounted for — CVar-α doesn't produce the smooth risk/return tradeoff the method is meant to deliver at this training budget.

Legend: mean held-out P&L (1 std, 4 seeds) (blue), CVar_{0.01} (1 std, 4 seeds) (pink).



AGENT	α	MEAN P&L	MEAN P&L (15 UNSEEN EP, P&R)	MEAN Δ	T-TEST P	WILCOXON P	SEEDS
QR-DQN	0.05	+630.819 $\pm$ +571.521	-880.400 $\pm$ +486.390	+0.696 $\pm$ +0.571	4		
QR-DQN	0.10	+577.337 $\pm$ +997.299	-1944.281 $\pm$ +2717.050	+1.577 $\pm$ +1.884	4		
QR-DQN	0.25	+999.651 $\pm$ +309.139	-445.192 $\pm$ +138.858	+3.893 $\pm$ +0.266	4		
QR-DQN	0.50	+384.968 $\pm$ +468.714	-23.275 $\pm$ +395.179	+1.267 $\pm$ +1.917	4		
QR-DQN	1.00	+326.284 $\pm$ +288.761	-505.121 $\pm$ +515.774	+1.651 $\pm$ +1.468	4		
IQN	0.05	-60.041 $\pm$ +215.008	-1501.388 $\pm$ +1746.026	+0.865 $\pm$ +0.268	4		
IQN	0.10	+167.542 $\pm$ +88.301	-968.822 $\pm$ +556.929	+0.278 $\pm$ +0.364	4		
IQN	0.25	+170.992 $\pm$ +729.278	-2559.782 $\pm$ +937.899	+0.116 $\pm$ +0.414	4		
IQN	0.50	+122.820 $\pm$ +73.164	-964.214 $\pm$ +1030.920	+0.542 $\pm$ +0.268	4		
IQN	1.00	-60.265 $\pm$ +582.986	-2540.835 $\pm$ +1414.460	+0.253 $\pm$ +0.734	4		

Encoder ablation — QR-DQN, normal regime

QR-DQN's held-out Sharpe using handcrafted features vs. two learned snapshot representations (CNN, autoencoder) — handcrafted wins decisively (+4.19 vs. +0.45 and +0.18, both p<0.0001).

Handcrafted

CNN

Autoencoder

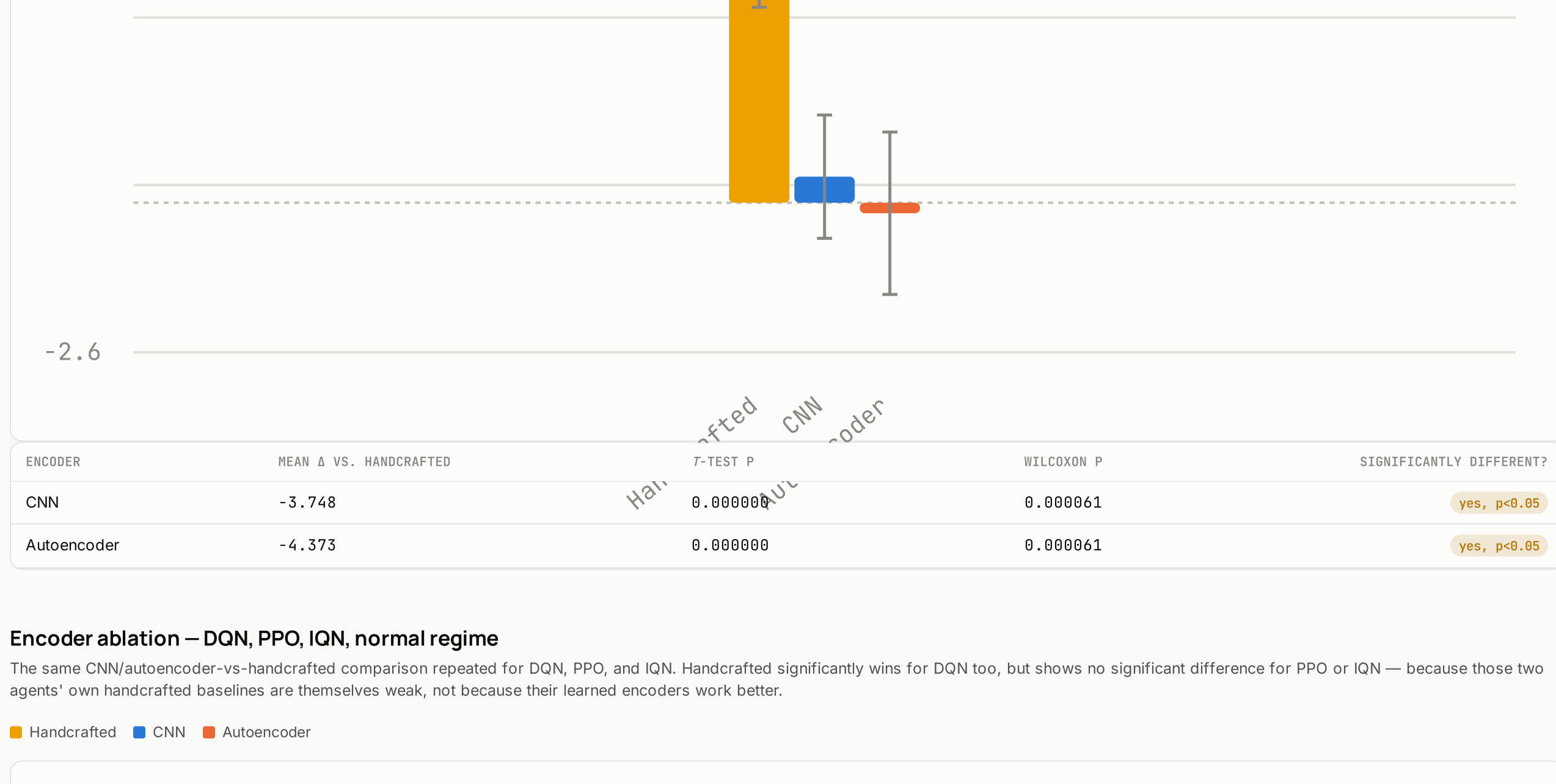
QR-DQN HELD-OUT SHARPE BY ENCODER

6.0

Encoder ablation – QR-DQN, normal regime

The same CNN+Autoencoder-vs.-handcrafted comparison repeated for DQN, PPO, and IQN. Handcrafted significantly wins for DQN too, but shows no significant difference for PPO or IQN — because those two agents' own handcrafted baselines are themselves weak, not because their learned encoders work better.

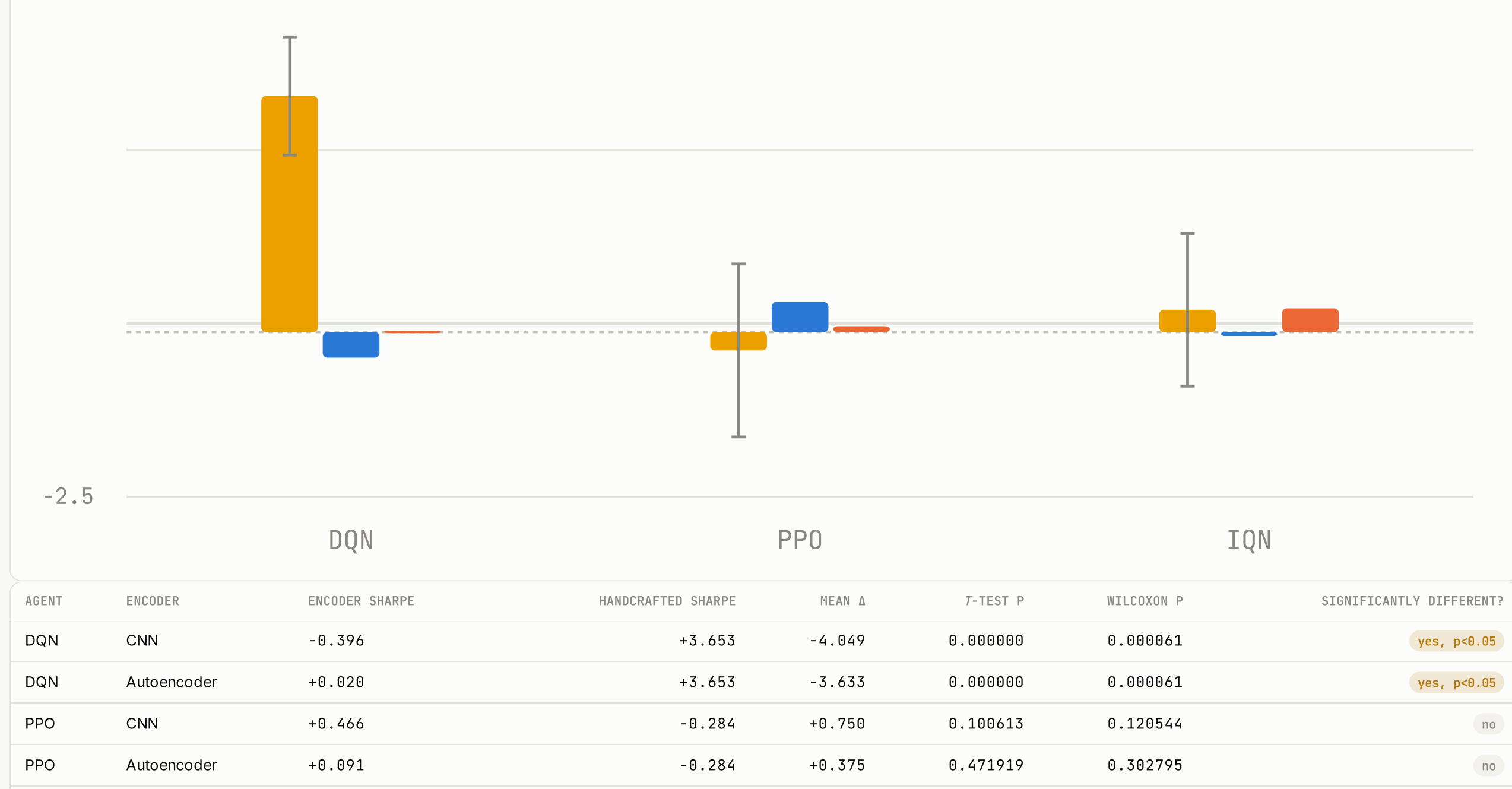
Legend: Handcrafted (blue), CNN (orange), Autoencoder (red).



Encoder ablation – DQN, PPO, IQN, normal regime

The same CNN+Autoencoder-vs.-handcrafted comparison repeated for DQN, PPO, and IQN. Handcrafted significantly wins for DQN too, but shows no significant difference for PPO or IQN — because those two agents' own handcrafted baselines are themselves weak, not because their learned encoders work better.

Legend: Handcrafted (blue), CNN (orange), Autoencoder (red).



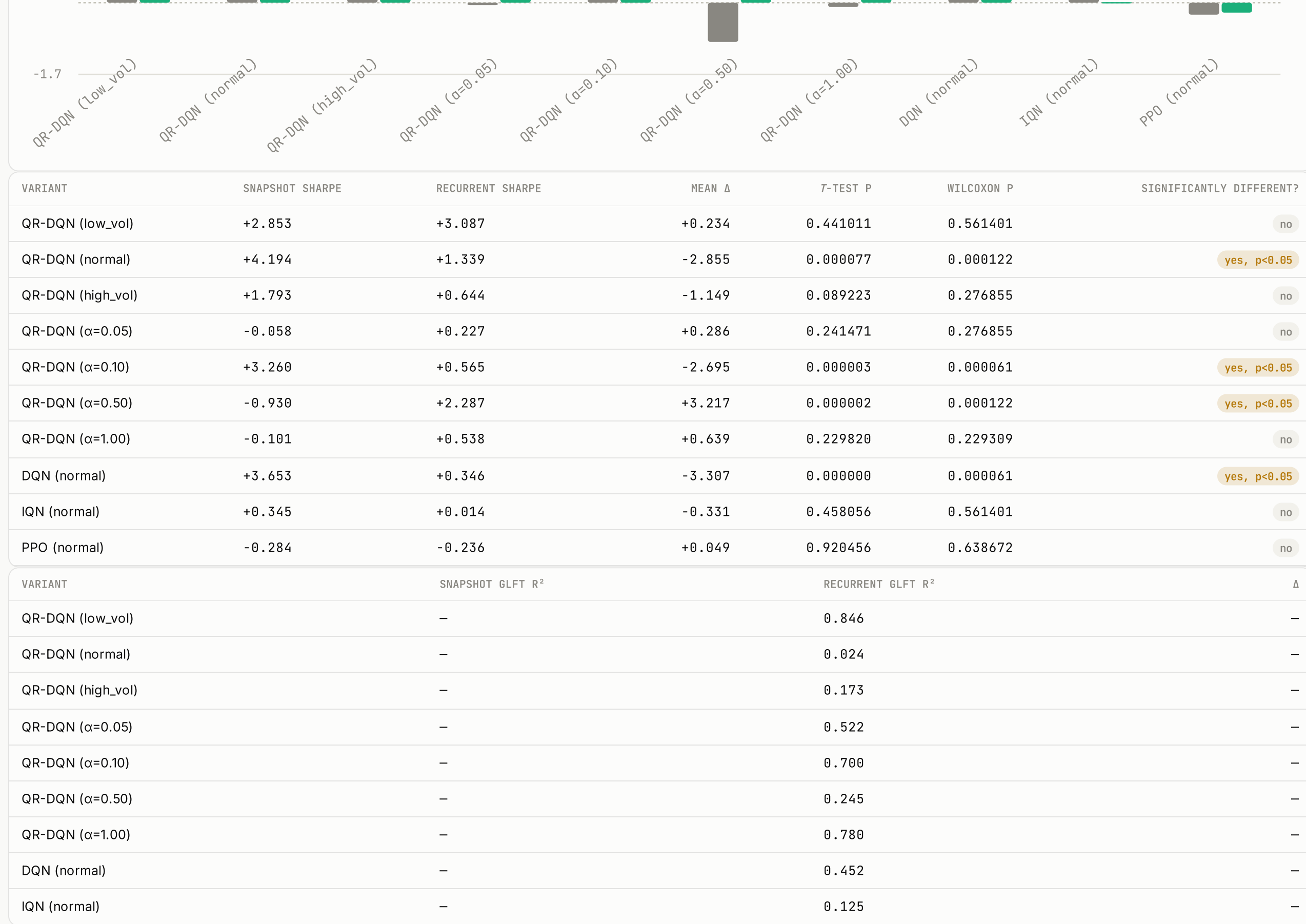
AGENT	ENCODER	ENCODER SHARPE	HANDCRAFTED SHARPE	MEAN Δ	T-TEST P	WILCOXON P	SIGNIFICANTLY DIFFERENT?
DQN	CNN	-0.396	+3.653	-0.234	0.000000	0.000061	yes, p<0.05
DQN	Autoencoder	+0.020	+3.653	-3.633	0.000000	0.000001	yes, p<0.05
PPO	CNN	+0.466	-0.284	+0.750	0.100613	0.120544	no
PPO	Autoencoder	-0.093	-0.284	+0.375	0.471919	0.302795	no
IQN	CNN	-0.061	+0.345	-0.406	0.455486	0.638672	no
IQN	Autoencoder	+0.366	+0.345	+0.021	0.963372	0.934082	no

AGENT	ENCODER	HANDCRAFTED GLTF R ²	ENCODER GLTF R ²	Δ
DQN	CNN	0.158	0.007	-0.151
DQN	Autoencoder	0.158	0.010	-0.148
PPO	CNN	—	0.063	—
PPO	Autoencoder	—	—	—
IQN	CNN	0.447	0.027	-0.420
IQN	Autoencoder	0.447	—	—

Recurrent vs. snapshot: does temporal memory help?

Each agent's standard snapshot-encoder policy vs. the same agent with an LSTM backbone in its place, across 10 agent/regime/CVar-risk combinations. Recurrent integration never significantly helps, significantly hurts QR-DQN and DQN in the normal regime, and its effect when combined with CVar risk-sensitivity is alpha-dependent rather than a consistent interaction.

Legend: snapshot (blue), recurrent (LSTM) (green).



VARIANT	SNAPSHOT SHARPE	RECURRENT SHARPE	MEAN Δ	T-TEST P	WILCOXON P	SIGNIFICANTLY DIFFERENT?
QR-DQN (low_vol)	+2.853	+3.087	+0.234	0.441011	0.561401	no
QR-DQN (normal)	+4.194	-2.855	-7.049	0.000077	0.000122	yes, p<0.05
QR-DQN (high_vol)	+1.793	+0.644	-1.149	0.089223	0.276855	no
QR-DQN (α=0.05)	-0.058	+0.227	+0.286	0.241471	0.276855	no
QR-DQN (α=0.10)	+3.260	+0.565	-2.695	0.000003	0.000061	yes, p<0.05
QR-DQN (α=0.50)	-0.930	+2.287	+3.217	0.000002	0.000122	yes, p<0.05
QR-DQN (α=1.00)	-0.101	+0.346	+0.639	0.229020	0.229020	no
DQN (normal)	+3.653	-0.854	-3.307	0.000000	0.000001	yes, p<0.05
IQN (normal)	+0.345	+0.014	-0.331	0.458056	0.561401	no
PPO (normal)	-0.284	-0.236	+0.049	0.920456	0.638672	no

VARIANT	SNAPSHOT GLTF R ²	RECURRENT GLTF R ²	Δ
QR-DQN (low_vol)	—	0.046	—
QR-DQN (normal)	—	0.024	—
QR-DQN (high_vol)	—	0.173	—
QR-DQN (α=0.05)	—	0.522	—
QR-DQN (α=0.10)	—	0.700	—
QR-DQN (α=0.50)	—	0.245	—
QR-DQN (α=1.00)	—	0.780	—
DQN (normal)	—	0.452	—
IQN (normal)	—	0.125	—

Full results table

Every figure charted above, in one place.

AGENT	TYPE	REGIME	EPISODES	STATUS	HELD-OUT SHARPE (15 UNSEEN EP, P&R)	± STD	PEAK EVAL SHARPE	EP	FINAL EVAL SHARPE (LAST 3, 10-SAMPLE)	TRAIN-TEST, MEAN (DIAGNOSTIC ONLY)	GLTF R ²	AS R ²
SARSA	RL	low_vol	375 / 500	early stopped	+0.583	0.94	+0.992	225	+0.248	+0.149	0.146	0.086
SARSA	RL	normal	425 / 500	early stopped	+0.114	1.31	+2.103	225	-0.777	+0.044	0.058	0.098
SARSA	RL	high_vol	300 / 500	early stopped	-1.184	1.48	-0.129	200	-1.100	-1.346	0.000	0.002
DQN	RL	low_vol	350 / 500	early stopped	+0.041	0.48	+0.356	25	-0.289	-0.080	0.021	0.077
DQN	RL	normal	500 / 500	complete	+3.653	0.91	+5.327	475	+0.749	+0.993	0.158	0.018
DQN	RL	high_vol	500 / 500	complete	+2.933	1.58	+5.772	375	+4.633	+1.751	0.449	0.589
PPO	RL	low_vol	500 / 500	complete	+0.219	0.74	+1.056	400	-0.029	-0.038	—	—
PPO	RL	normal	400 / 500	early stopped	-0.284	1.34	+1.535	225	-0.228	-0.132	—	—
PPO	RL	high_vol	500 / 500	complete	-1.419	1.31	+0.008	300	-1.497	-1.275	—	—
QR-DQN	RL	low_vol	500 / 500	complete	+2.853	0.79	+2.945	325	+0.854	+0.429	0.508	0.788
QR-DQN	RL	normal	450 / 500	early stopped	+4.194	0.85	+5.483	275	-5.703	+1.083	0.304	0.430
QR-DQN	RL	high_vol	475 / 500	early stopped	+1.793	1.68	+3.307	325	+2.224	+0.328	0.347	0.364
IQN	RL	low_vol	500 / 500	complete	+0.434	0.68	+1.613	250	+0.854	+0.364	0.983	0.641
IQN	RL	normal	400 / 500	early stopped	+0.345	1.10	+1.796	250	-0.741	+0.117	0.447	0.563
IQN	RL	high_vol	325 / 500	early stopped	-0.054	1.86	+0.877	125	-0.357	-0.618	0.262	0.379
Fixed-Spread	baseline	low_vol	50 / 50	n/a (I.I.D.)	+0.944	0.65	—	—	—	+0.959	—	—
Fixed-Spread	baseline	normal	50 / 50	n/a (I.I.D.)	-1.398	1.81	—	—	—	+0.679	—	—
Fixed-Spread	baseline	high_vol	50 / 50	n/a (I.I.D.)	-2.179	1.19	—	—	—	-1.961	—	—
Avellaneda-Stokov	baseline	low_vol	50 / 50	n/a (I.I.D.)	+1.241	1.01	—	—	—	+1.156	—	—
Avellaneda-Stokov	baseline	normal	50 / 50	n/a (I.I.D.)	-0.901	1.03	—	—	—	-0.999	—	—
Avellaneda-Stokov	baseline	high_vol	50 / 50	n/a (I.I.D.)	-2.719	1.19	—	—	—	-1.831	—	—
GLTF	baseline	low_vol	50 / 50	n/a (I.I.D.)	+1.240	1.00	—	—	—	+1.302	—	—
GLTF	baseline	normal	50 / 50	n/a (I.I.D.)	-1.165	1.05	—	—	—	-0.992	—	—
GLTF	baseline	high_vol	50 / 50	n/a (I.I.D.)	-2.179	1.19	—	—	—	-2.368	—	—